SUBJECT CODE : 23A01702b

RAILWAYS, AIRPORTS, DOCKS AND HARBOUR ENGINEERING

UNIT – III : AIRPORT ENGINEERING

(Handwritten Study Notes - Exam Oriented)

Syllabus (as given):

Airport Site Selection – Factors Affecting Site Selection and Surveys – Runway Orientation – Wind Rose Diagram – Basic Runway Length – Correction for Runway Length – Terminal Area – Layout and Functions – Concepts of Terminal Building – Simple Building, Linear Concept, Pier Concept and Satellite Concept – Typical Layouts

1. Introduction to Airport Engineering

An **airport (aerodrome)** is an area of land (or water) with buildings and facilities for the **arrival, departure and ground movement of aircraft**, and for handling passengers and cargo. Air transport is the **fastest** mode for long-distance travel and is vital for trade, tourism and defence.

- **ICAO** – International Civil Aviation Organisation (standards in Annex 14, Aerodromes). **DGCA** – regulator in India; **AAI** – Airports Authority of India builds/operates most airports.
- **Airside**: runways, taxiways, aprons, hangars, control tower, navigational aids. **Landside**: terminal building, roads, car parks, ground transport links.

Classification of airports (ICAO aerodrome reference code)

Code no.	Reference field length	Code letter	Wing span
1	< 800 m	A	< 15 m
2	800 – 1200 m	B	15 – 24 m
3	1200 – 1800 m	C	24 – 36 m
4	≥ 1800 m	D / E / F	36–52 / 52–65 / 65–80 m

- By use: international, domestic, military, STOL / heliport, general aviation; by size and traffic: Category I, II, III etc.

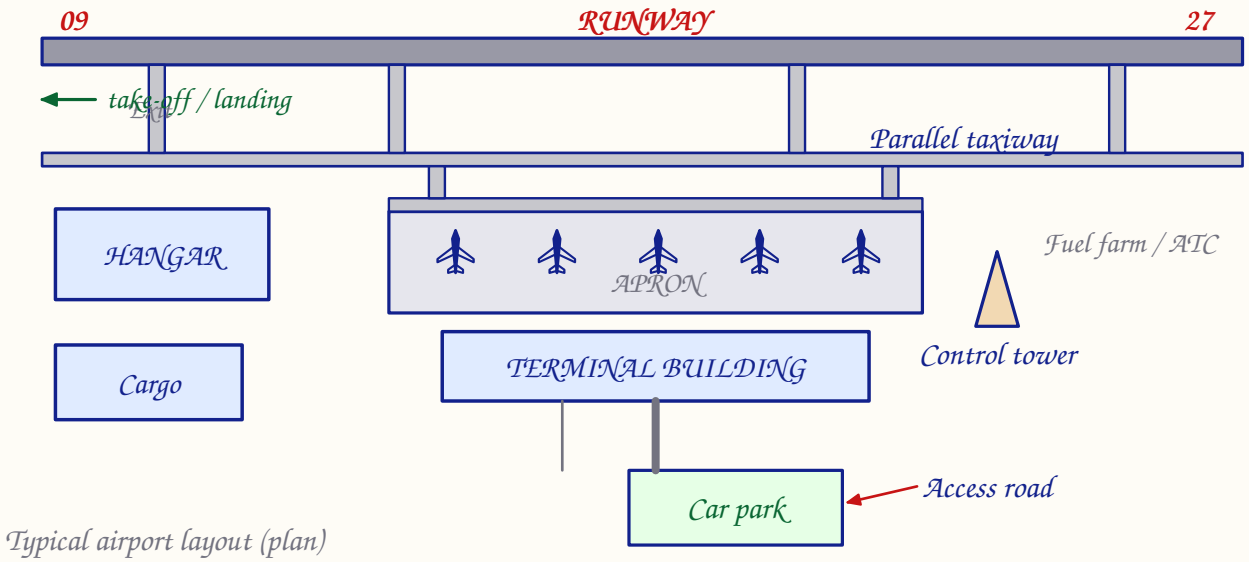


Fig: Typical airport layout

2. Airport Site Selection

Selecting the site is the first and most important step because an airport is costly, permanent and affects a large area. A site is chosen after comparing several alternative sites.

Factors affecting site selection

Factor	Consideration
Regional plan	Airport should fit into the regional / national air-transport plan and the development of the region.
Airport use & traffic	Type of operations (civil, military, cargo), type and number of aircraft, future growth.
Proximity to other airports	Avoid air-space conflict; sufficient separation from other aerodromes.
Ground accessibility	Near the city but not too close; good road/rail links; travel time ≈ 30-45 min.
Topography	Flat, well-drained land; less earthwork; good natural approach funnels; no hills.
Obstructions	Clear approach zones – no tall buildings, towers, chimneys, hills or power lines.

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Factor	Consideration
Surrounding land use	Avoid residential areas (noise); Industries, hospitals, schools away from approach.
Noise & environment	Aircraft noise, air pollution, bird activity (landfills, ponds) must be avoided.
Meteorological	Wind direction & speed, visibility (fog, smoke), rainfall, temperature, cloud height.
Soil & drainage	Good bearing soil, high water table avoided; natural drainage; low cost of pavement.
Utilities	Water, electricity, sewage, telecom available at reasonable cost.
Future expansion	Land available for extra runway, terminal & cargo; Economy of land acquisition.

Surveys for airport site

- Topographic / contour survey of the site and surroundings; aerial photographs & Google imagery.
- Soil survey – bearing capacity, CBR, sub-soil water; hydrological survey – rainfall, drainage outfall.
- Meteorological survey – wind data for 5-10 years, visibility, temperature, rainfall → wind rose.
- Traffic & economic survey – passengers, aircraft movements, cargo, catchment area; environmental impact assessment (EIA).
- Obstruction survey – check the obstacle limitation surfaces for height of structures and trees.

Obstacle Limitation Surfaces (OLS) – ICAO (Code 4)

Surface	Slope / size
Approach surface	starts at 60 m from threshold; slope 2 % (1 in 50) for first section
Transitional surface	slope 14.3 % (1 in 7) from strip edge and approach edge up to inner horizontal surface
Inner horizontal surface	45 m above aerodrome elevation, radius about 4000 m

Surface	Slope / size
Conical surface	slope 5 % (1 in 20) outwards from inner horizontal, up to 100 m height

3. Runway Orientation and Wind Rose

Aircraft take off and land against the wind (head wind) because it shortens the run and gives stability. So the runway should be oriented along the direction of the prevailing wind, so that the cross-wind component is minimum.

$$\text{Cross-wind component} = V \cdot \sin\theta$$

V = wind speed, θ = angle between the wind and the runway axis. Ex: $V = 40$ km/h, $\theta = 30°$ → cross wind = $40 \times \sin 30° = 20$ km/h; head wind = $40 \cos 30° = 34.6$ km/h.

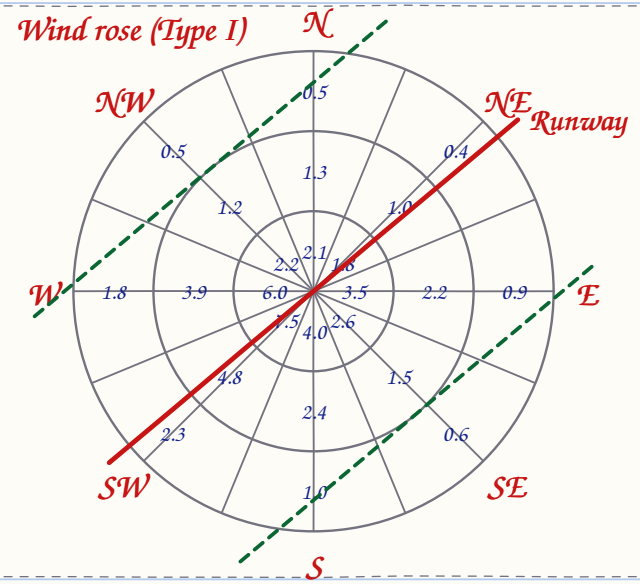
Reference field length	Max permissible cross-wind (ICAO)
≥ 1500 m (Code 4 etc.)	20 knots = 37 km/h
1200 m to 1500 m	13 knots = 24 km/h
< 1200 m	10 knots = 19 km/h

- Wind coverage: percentage of time during which the cross-wind is within the limit. The runway should give at least 95 % coverage; if not, an additional cross-wind runway is needed.

Wind rose diagram

A wind rose is a graphical record of wind direction, intensity and duration at the site. Data are collected for at least 5 – 10 years.

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Use of wind rose

1. Draw 16 radial lines; mark % of time wind blows from each direction (in speed ranges).
2. Take a transparent strip: centre line = runway, side lines at ± permissible cross-wind.
3. Place the strip at the centre of rose and rotate.
4. Add all % lying inside the strip (wind coverage).
5. The position with max total gives best orientation; coverage should be ≥ 95 %.

Red: centre line Green: ± cross-wind limit

ring = wind speed; figures = % of time (illustrative)

Fig: Wind rose with the runway template

Type	Description
Type I	Shows wind direction and intensity (speed). 16 radial lines; numbers on each radial give % of time for different speed ranges. Template of 3 parallel lines is rotated.
Type II	Shows direction, intensity and duration (frequency). Wind speeds in concentric zones of the 8 or 16 directions; used by rotating the template and adding the %.

- Steps (Type I): collect wind data → draw rose → make a transparent strip (centre line + two lines at ± cross-wind limit) → rotate about centre → select the orientation with maximum total % inside the strip.
- Other factors: terrain, obstruction in approach, noise, bird hazard, adjacent airport air space.
- Runway numbering: magnetic azimuth ÷ 10 rounded to the nearest whole; 09 / 27 = 90° and 270°. Parallel runways add L, C, R.

4. Basic Runway Length

Basic runway length is the length required for an aircraft to take off or land under standard conditions, taken from the manufacturer's performance curves, for the critical (most demanding) aircraft.

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Standard conditions

- Airport at mean sea level; temperature 15°C (standard atmosphere).
- Runway level (zero gradient); no wind; maximum take-off weight; dry & hard surface.
- No wind, no stopway or clearway; no obstacles; no engine failure.

Corrections to basic runway length (L)

Correction	Rule
1. Elevation	Increase 7 % per 300 m above MSL (air is thinner): $L2 = L\,(1 + 0.07\,h / 300)$
2. Temperature	Increase 1 % for every 1°C that the Airport Reference Temperature (ART) exceeds the standard atmospheric temperature at that elevation: $L3 = L2\,[1 + 0.01\,(ART - Ts)]$
Combined limit	Elevation + temperature correction should not exceed 35 % of L
3. Gradient	ICAO: 20 % for each 1 % effective gradient: $L4 = L3\,(1 + 0.2\,G)$

$$Ts = 15 - 0.0065\,h\,; \quad ART = Ta + (Tm - Ta)/3$$

- h = elevation of the airport in m; Ta = monthly mean of the average daily temperature of the hottest month; Tm = monthly mean of maximum daily temperature of the same month.
- Effective gradient = (max. difference in elevation between highest and lowest point on the runway centre line) ÷ runway length × 100.

Example 1 – Runway length

Basic length = 1800 m, airport elevation = 270 m, ART = 29°C, effective gradient = 0.5 %.

(a) Elevation: $L2 = 1800 \times (1 + 0.07 \times 270/300) = 1800 \times 1.063 =$ **1913.4 m**.

(b) $Ts = 15 - 0.0065 \times 270 = 13.245$°C; $ART - Ts = 15.755$°C → $L3 = 1913.4 \times 1.15755 =$ **2214.9 m**.

Check: 2214.9 / 1800 = 1.23 < 1.35 (OK, limit 35 %).

(c) Gradient: 20 % × 0.5 = 10 % → $L4 = 2214.9 \times 1.10 =$ **2436.4 m ≈ 2440 m**.

Example 2 – Airport Reference Temperature

Ta = 24°C, Tm = 32°C → ART = 24 + (32 – 24)/3 = **26.67°C**.

- Aircraft with higher weight/payload need longer runway; take-off length is usually more than landing length for jet aircraft.

5. Terminal Area

Terminal area is the portion of airport where passengers and cargo are transferred between ground transport and aircraft. It includes the terminal building, apron, car parking, access roads and the support buildings.

Components of the terminal area

- Terminal building – passenger processing: check-in, security, immigration/customs, baggage claim, lounges, shops.
- Apron – place for aircraft to park, load/unload, refuel; with gate positions.
- Hangars (maintenance), control tower, fire station, fuel farm, cargo terminal, and car park / roads.

Functions of terminal area

- Changing mode: ground → air; processing passengers, baggage and cargo; information & communication.
- Providing safe and fast movement of passengers with minimum walking distance; comfort and services.

Aircraft parking configurations on the apron

Type	Remarks
Nose-in	Aircraft parked at 90° to building; nose towards terminal; uses less space, needs tug to push-back
Angled nose-in	Aircraft at an angle; can move out under own power; more apron space
Angled nose-out	Nose pointing away from building at angle
Parallel	Parallel to terminal; needs more frontage; easy movement

Number of gates

$$G = V \cdot T / U$$

G = number of gates, V = design hourly volume (aircraft/hr), T = gate occupancy time (hr), U = utilisation factor (0.6 – 0.8). Ex: $V = 8$, $T = 45$ min $= 0.75$ hr, $U = 0.8 \rightarrow G = 8 \times 0.75 / 0.8 = 7.5 \approx$ 8 gates.

6. Concepts of Terminal Building

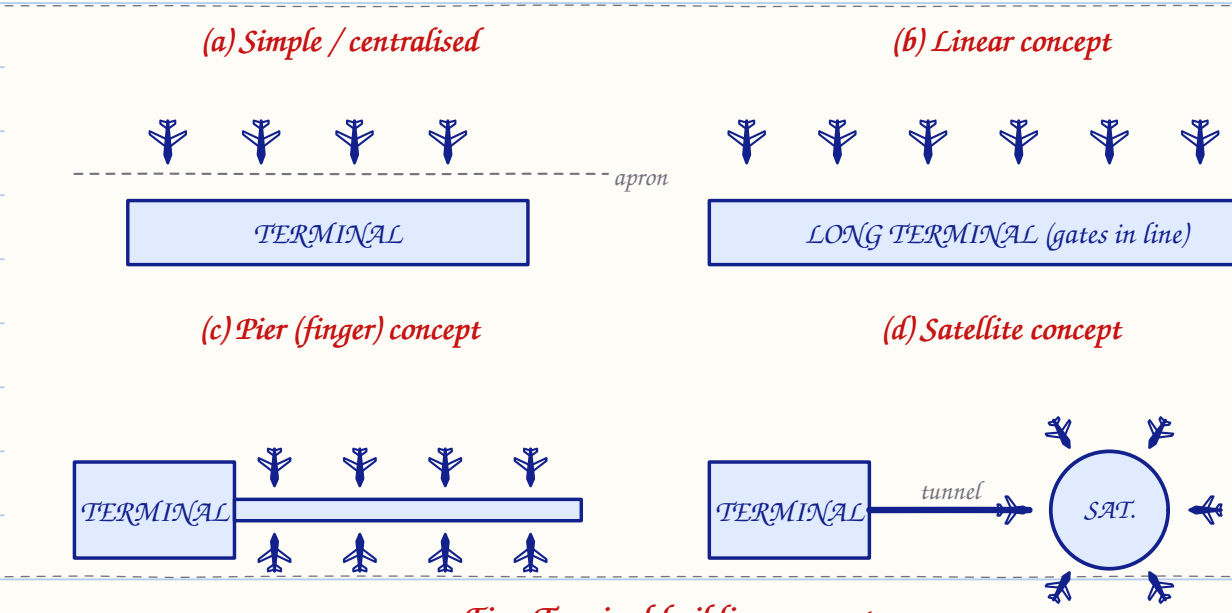


Fig: Terminal building concepts

Concept	Description	Merits	Demerits
Simple (centralised)	Passenger lounge and aircraft close to the single building; aircraft parked on the apron in front	Simple, cheap, short walking, for small airports	Limited expansion; apron crowded
Linear	Building extended linearly; aircraft parked along its length; gates in a line	Short walking, easy expansion in length, direct access	Long distance for transfer if many gates; large frontage
Pier (finger)	A long pier (finger) projects from the terminal; aircraft parked on both sides of pier	More gates on small land; protected passenger walkway	Long walking; aircraft manoeuvre restricted
Satellite	Detached circular/ square satellite building connected to main terminal by underground/ elevated tunnel; aircraft around satellite	Many gates compact; automatic people mover; flexible	Costly; baggage handling complex; long travel for transfer

- Other concept: Transporter (remote apron) – aircraft parked away; passengers carried by buses / mobile lounges. Cheap but less convenient.

7. Typical Layouts of Runways and Airports

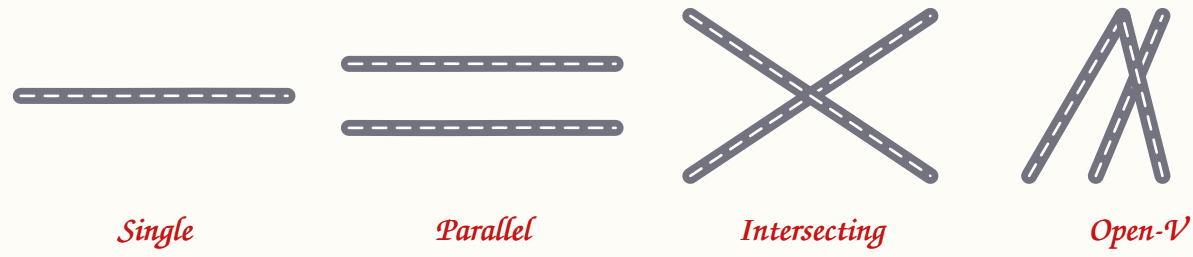


Fig: Runway configurations

- **Single runway** – simplest; capacity limited; used in small airports; **parallel runways** – two runways at a distance; capacity greatly raised (simultaneous landing/take-off).
- **Intersecting runways** – two runways cross; used where winds from different directions; the capacity depends on the crossing point (near the thresholds is better).
- **Open-V runways** – two diverging runways; when wind comes from the closed end, both can be used simultaneously, giving greater capacity.
- Taxiways connect runways with apron; **parallel taxiway** and rapid exit taxiways increase capacity and reduce runway occupancy time.

8. Declared Distances and Runway Terms

Term	Meaning
TORA	Take-Off Run Available – length of runway declared available for ground run of an aircraft taking off
TODA	Take-Off Distance Available = TORA + clearway
ASDA	Accelerate-Stop Distance Available = TORA + stopway
LDA	Landing Distance Available – runway length usable for landing (from displaced threshold)
Stopway	Area beyond take-off run, as wide as runway, able to support an aircraft during an abandoned take-off
Clearway	Obstacle-free area beyond the runway end under control of authority, width ≥ 150 m, slope up to 1.25 %

Term	Meaning
Threshold	Beginning of the usable landing portion of the runway; can be displaced due to obstacles

- Runway length is checked against balanced field length – the distance needed to either continue take-off or stop after an engine failure at the decision speed V1.

9. Wind Coverage – Worked Example

Example 3 – Orientation of a runway

A wind rose (Type I) has been drawn for a site; runway length < 1200 m, so permitted cross-wind = 19 km/h.

Template placed with centre line along N–S (0°): total % of winds inside the strip = 89.0 % (illustrative values).

Rotated to N 20° E: total inside strip = 94.5 %; rotated to N 35° E: 96.2 %; N 50° E: 93.0 %.

Maximum coverage = 96.2 % (> 95 %) → runway orientation: N 35° E (bearing 35° / 215° → number 04 / 22).

Example 4 – Cross-wind check

Wind from 60° to runway at 45 km/h: cross wind = 45 sin 60° = 38.97 km/h (> 37) → not permissible for a Code 4 runway;

the same wind at 40° gives 45 sin 40° = 28.9 km/h (< 37 km/h) → permissible.

- Runways are numbered by magnetic azimuth: a 35° bearing → '04' (rounded 3.5 → 4) at one end and '22' at the other end (215° → 21.5 → 22).

10. Terminal Planning Considerations

Passenger flow in a terminal

Departing: Kerb / car park → check-in (baggage drop) → security → immigration (if international) → departure lounge / shops → boarding gate → aircraft.

Arriving: Aircraft → arrival gate → immigration / health → baggage claim → customs → arrival hall → kerb / transport.

Planning principles

- Short and simple passenger and baggage flow; **separate** departing and arriving flows; clear signs.
- Flexibility to **expand** in modules; segregation of domestic/international and of passengers/cargo.
- Adequate **kerb length**, car parking and a good public transport link.
- Terminal area size is based on **peak-hour passengers** (design busy-hour) and the level of service wanted.

Apron and hangars

- **Apron** pavement is strong (rigid pavement preferred because of fuel spill & heavy load); slope about 0.5 – 1 % for drainage.
- **Hangars**: for storing and maintenance of aircraft; located away from runways and near apron; sized by wing span and tail height; types – arched, cantilever, portal frame.
- **Control tower**: visual control of aerodrome; located to see all runways, taxiways and the apron; Radar (ATC) provides airspace control.

Aircraft stand clearance (concept)

- Wing-tip clearance at a stand: 3 m (code C), 4.5 m (D, E) and 7.5 m (F) as per ICAO, so apron width and gate spacing depend on the **wing span** of the largest aircraft.

11. Comparison and Revision Notes

Point	Simple	Linear	Pier	Satellite
Walking distance	Shortest	Short	Long	Longest
Gates capacity	Low	Moderate	High	Very high
Expansion	Difficult	Easy (length)	Easy (pier)	Easy (new satellite)
Cost	Lowest	Low	Moderate	Highest
Example use	Small airports	Medium airports	Large airports	Major hubs

One-line facts

- Aircraft take-off/landing into the wind; wind coverage ≥ 95 %.
- ICAO cross wind limits: 37 km/h (≥1500 m), 24 km/h (1200–1500 m), 19 km/h (< 1200 m).
- Basic length corrections: elevation → temperature → gradient (in this order).
- ART = Ta + (Tm – Ta)/3; Ts = 15 – 0.0065 h.
- Terminal concept: Simple, Linear, Pier, Satellite (and Transporter).

Important Questions – Unit III

Short answer (2 marks)

- List any four factors affecting airport site selection.
- What is a wind rose? Mention the types.
- State the ICAO limits of cross-wind component.
- What are the standard conditions for basic runway length?
- What is the apron? Name the terminal building concepts.

Long answer (10 marks)

- Discuss the factors that govern the selection of a site for an airport; add notes on surveys.
- Explain the Type I wind rose and how the orientation of runway is found.
- Explain the corrections applied to the basic runway length with a numerical example.
- Explain the layout of the terminal area and its functions.
- Sketch and explain the simple, linear, pier and satellite concepts of terminal building.

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Quick revision: Elevation +7 %/300 m | Temp +1 %/°C over standard | Max total 35 % | Gradient +20 % per 1 % | Cross wind 37/24/19 km/h | Coverage ≥ 95 % | $G = VT/U$.